Al-Ahadeeth: A Visualization Tool of the Hadiths' Chain of Narrators

Hamada R. H. Al-Absi[1](✉), Devi G. Kurup[1], Amina Daoud[1], Jens Schneider[1], Wajdi Zaghouani[2], Saeed Mohd H. M. Al Marri[3], and Younss Ait Mou[1](✉)

[1] College of Science and Engineering, Hamad Bin Khalifa University, Doha, Qatar
{haalabsi,dkurup,aminadaoud,jeschneider,ymou}@hbku.edu.qa

[2] College of Humanities and Social Sciences, Hamad Bin Khalifa University, Doha, Qatar
wzaghouani@hbku.edu.qa

[3] College of Shariah and Islamic Studies, Qatar University, Doha, Qatar
saealmarri@qu.edu.qa

Abstract. Hadiths are the records of words, actions, and silent approval attributed to the Prophet Mohammed (PBUH). Their significance is second only to the Holy Quran within the Islamic faith. They have been collected, curated, and published by the most respected Islamic scholars through time. Each Hadith begins with a chain of narrators (sanad), distinguishing them from the main text (Matan), serving as a means for Islamic scholars to assess their strength and veracity. Most people read the approved six books of Hadiths, often focusing solely on the content (Matan) or browsing through topics, neglecting the profound intricacies hidden within the individual narrators' backgrounds. Furthermore, the data is accessed without a visual user interface, imposing limitations due to the purely text-based nature. To alleviate this, we present Al-Ahadeeth, a visualization tool of the chain of narrators of Hadiths. Mapping narrators and narrated-to relations to nodes and edges in a directed graph, we use node size and edge thickness to show occurrence frequencies. Using the concept of generations, we place nodes on concentric rings based on the topological index to compute a 2D layout. Due to the high degree of interconnection in the community of narrators, traditional graph embedding techniques such as multi-dimensional scaling (MDS), t-distributed stochastic neighbor embedding (t-SNE) or uniform manifold approximation and projection (UMAP) tend to map the entire graph to a single cluster and give unsatisfactory results. In contrast, our approach uses the concept of generations to compute the layout of the graph which places nodes on concentric rings to indicate the generation.

Keywords: Hadith · Chain of Narrators · Directed Graph · Visualization

© The Author(s), under exclusive license to Springer Nature Switzerland AG 2024
C. Stephanidis et al. (Eds.): HCII 2024, CCIS 2117, pp. 3–8, 2024.
https://doi.org/10.1007/978-3-031-61953-3_1

1 Introduction

In Islam, Hadiths (singular: Hadith; in Arabic: حديث) are the words, actions, silent approvals, and habits of Prophet Mohammed-Peace Be Upon Him (PBUH). They are considered the second source of Islamic teachings following the Holy Quran [1,2]. In Arabic, the word "Hadith" in its singular pronunciation form means a "speech" or "report" and its plural form is "Ahadeeth" or "Hadiths" and refers to collections. Scholars have collected, investigated, curated, and published these Hadiths over time, adhering to a strict methodology to ensure the authenticity of each Hadith [3]. An incredibly rich collection of historical information is found in the Kutub al-Sittah—The Six Books. These collections include commentary on the works of Sahih Bukhari, Sahih Muslim, Sunan Abu Dawood, Jami al-Tirmidhi, Sunan al-Sughra, and Sunan ibn Majah and/or Muwatta Malik. In addition to the Hadiths themselves, geographical data and narrators' biographies have likewise been collected and compiled into books such as "The Great History" by Bukhari or al-Jarh wa al-Ta'dil by Abi Hatim, to give just two examples. The usage of this data varies greatly depending on the objective. A Hadith comprises two parts: the *"Matn"* and the *"Sanad"*. The *"Matn"* is the body of the Hadith that reports the action or reaction to specific situations, or teachings of the prophet PBUH, and the *"Sanad"* is the chain of narrators who have transferred the Hadith generation after another from *"Al Sahabah"* the companions of the Prophet PBUH, who passed these Hadiths to their successors, who are called *"Al Tabiun"*, and this was continued from one generation to another [2].

In the science of Hadith, validating each Hadith's authenticity is of significant importance. Scholars use the *"Sanad"* to validate Hadiths, a crucial step in preserving the teachings of Prophet Mohammed PBUH. A key criterion in this validation process is examining the narrators' character and the continuity of their chain [4]. Through this process, scholars have examined the trustworthiness and ethical conduct of narrators, categorizing them based on their reliability, and scrutinizing their connections to predecessors and whether these connections have continued to the Prophet PBUH or any of his companions, *"Al Sahabah"*, to establish the Hadiths' authenticity [4]. A major challenge for scholars studying the chain of narrators is the textual nature of the data. Apart from books, scholars utilize both online resources and offline databases, which offer vast and complex collections of literature. This constraint can impair both the accessibility and usability of Hadith data for authentication purposes, underscoring the necessity for improved tools and interfaces. Such advancements would enable more efficient analysis and validation of the chains of narration within Hadith literature. By overcoming these technological barriers, researchers could significantly enhance validation processes, ensuring the preservation and accurate interpretation of Prophetic traditions in contemporary scholarship.

Some researchers have used technological tools to study Hadith narrators, Saeed et al. [5] investigated the construction of narrative networks from the Sahih Muslim book. Using social network analysis techniques, the researchers aimed to discover primary narrators, interaction patterns, and network structural features. The study offered a technique for evaluating narrators and created a tool for narrative network analysis to help scholars and historians comprehend the

transmission of hadith. The research identified major narrators using centrality metrics and compared the findings to a comparable investigation on the Sahih Bukhari book. The comparison revealed both parallels and variations in the rankings of renowned narrators across the two surveys. The study emphasizes the possibility for future research to better explore the narrative network. Shukur et al. [6] introduced the Chain of Hadith Narrators Visualizer (CHN), a prototype designed to visually display the chains of narrators in Hadith literature using Information Visualization methods. Preliminary testing with 20 Islamic education students revealed favorable results, suggesting the efficiency of visual tools in assisting Hadith study. Another article by Siddiqui et al. [7] presented a system to automatically retrieve the Hadith chain of narrators using Named Entity Recognition (NER) and classification, allowing them to construct a network to visualize the narrator's chain. They used machine learning trained on an annotated Hadith corpus to identify the names of narrators. One common issue with the above studies, especially when visualizing the chain of narrators is the amount of clutter in the visualizations making them hard to navigate and understand or clearly find connections. Furthermore, traditional graph embedding techniques, such as multi-dimensional scaling (MDS) [8], t-distributed stochastic neighbor embedding (t-SNE) [9], or uniform manifold approximation and projection (UMAP) [10], often result in unsatisfactory results due to the high degree of interconnection among narrators.

The traditional text-based approach to studying Hadith chains of narrators poses significant challenges in comprehending the intricate relationships and patterns within the narration network. Hence, in this paper, we present our Al-Ahadeeth tool which employs the concept of generations to offer a visualization of the chain of narrators that is both apparent and readily accessible. Our tool aims to address these limitations by providing a visual representation and analysis tool that leverages the power of information visualization techniques. By mapping the complex network of narrators as an interactive graph, Al-Ahadeeth enables users to explore the connectivity patterns, identify central figures, and uncover insights that may be obscured in the textual format, making the complex connections of narrators more understandable and visually engaging.

2 Methodology

2.1 Data

We developed this tool using data from the Muslim Scholar Database [11]. Initially, we focused primarily on adding Sahih Bukhari data to the current prototype. This decision was made strategically, with the goal of refining our approach on a well-defined subset of data before extending to include all six Hadith books. Each narrator in the database is identifiable uniquely by a Muslim Scholar ID, allowing for a direct link to a wide range of metadata. This includes, but is not limited to, biographical and geographical data, which increase the tool's usability and depth of understanding.

The data obtained from the Muslim Scholar Database underwent a thorough preprocessing and cleaning process to ensure its suitability for integration

into the Al-Ahadeeth tool. This included handling missing or inconsistent data, resolving any conflicts or duplicates, and standardizing the formatting and representation of various attributes.

2.2 Tool Development

Our tool comes in two versions, a web version (https://www.hbku.edu.qa/cse/al-ahadeeth/) and iPad version (https://apps.apple.com/me/app/alahadeeth/id1590736786). To build both versions, we have leveraged the capabilities of a recent UI framework developed, and maintained by Google called **Flutter**. We have chosen this framework among others for its rapid live updates (hot-reload) and cross-platform capabilities. Therefore, a single source code was required to build and deploy both versions. This allowed us to focus on the research aspect of the project rather than on the maintenance of multiple source codes. We have used a CustomPainter to render the graph elements (nodes, edges, generation circles). We have opted for Flutter's Impeller runtime rendering capabilities to improve the rendering performances. That was particularly the case for situations where a large number of nodes, and edges had to be rendered simultaneously.

3 Al-Ahadeeth Tool

The Al-Ahadeeth tool enables users to visually explore the *"Sanad"*. Mapping narrators and narrated-to relations to nodes and edges in a directed graph, we

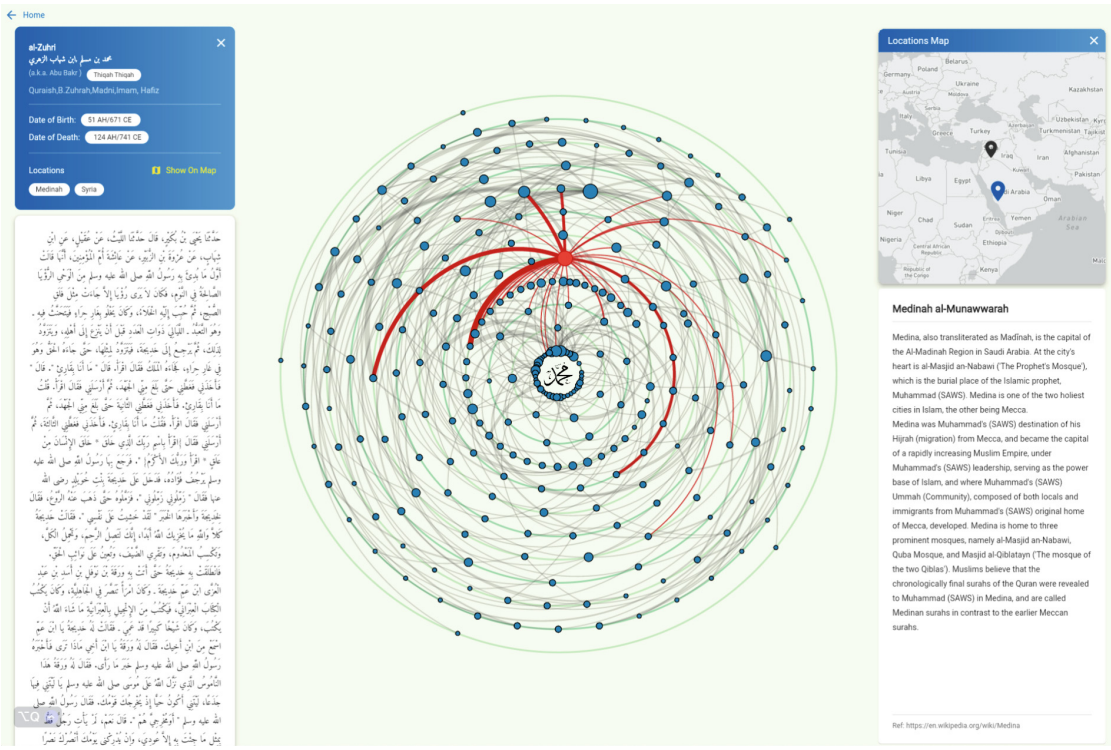

Fig. 1. Al-Ahadeeth tool Interface. Lines in red show the connection between one narrator and others. (Color figure online)

used node size and edge thickness to illustrate occurrence frequencies. Using the concept of generations, we placed nodes on concentric rings based on the topological index and used mass-spring relaxation to avoid edge intersections to compute our 2D layout. The tool provides the full text of the Hadiths (English and Arabic), geolocations of narrators, and short bios. Links to external online sources provide more detailed biographies. Figures 1 and 2 show examples of the tool. The Al-Ahadeeth tool offers a range of user interactions and features to facilitate exploration and analysis of the Hadith chains of narrators. Users can pan and zoom the graph view, enabling them to focus on specific regions or zoom out to observe the overall structure. Additionally, interactive filtering and search capabilities allow users to isolate narrators based on various criteria, such as biographical attributes, geographical regions, or occurrence frequencies. To enhance the user experience and provide contextual information, the tool displays relevant metadata associated with each narrator when hovering over or selecting a node. This includes biographical details, geographical locations, and links to external resources for further exploration. Furthermore, the tool supports the display of the full text of Hadiths, both in Arabic and English translations, allowing users to navigate between the visual representation and the textual content.

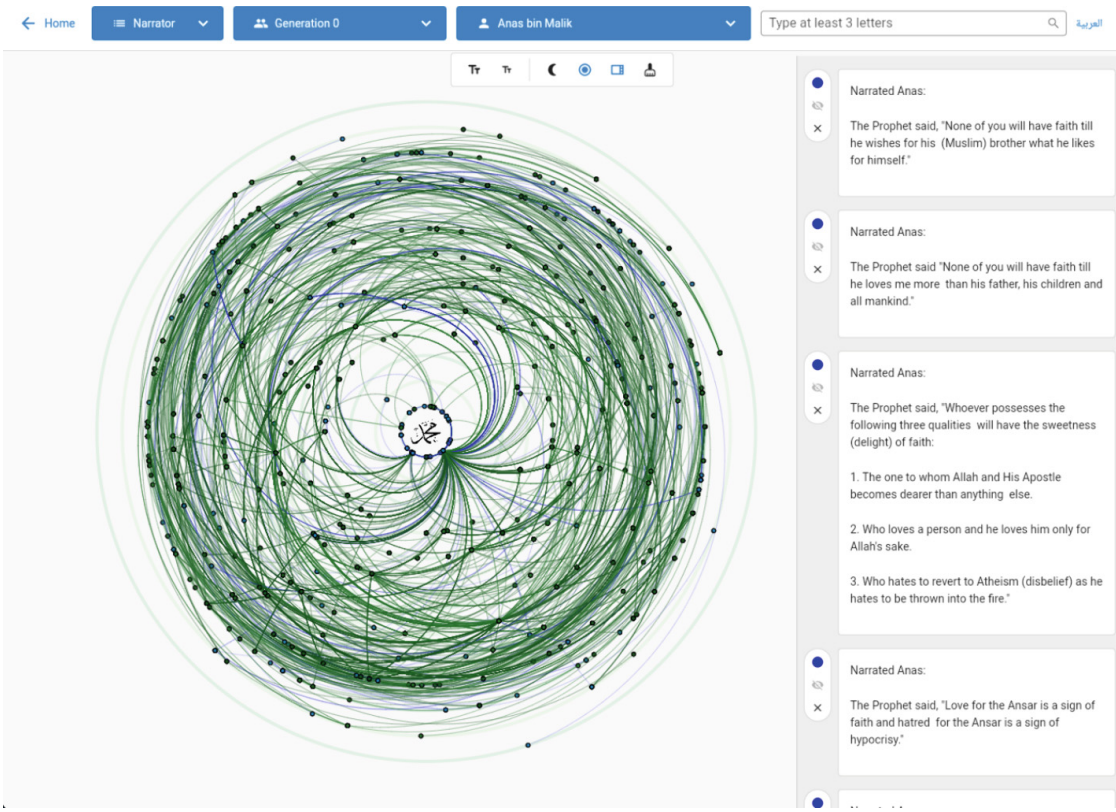

Fig. 2. An example of the Mobile application output when selecting one Narrator (Anas bin Malik). In the middle, the connection between Anas bin Malik and the other narrators is shown. On the right panel, the Hadiths are displayed.

4 Conclusion

In this paper, we presented Al-Ahadeeth, a visualization tool for the chain of narrators, *"Sanad"*, in Hadith literature. Using data from the Muslim Scholar Database, we focused on Sahih Bukhari for the current prototype, intending to eventually expand to include all six Hadith books. Al-Ahadeeth takes an innovative technique, mapping narrators and their relationships onto a directed graph, with node size and edge thickness expressing occurrence frequencies, and nodes positioned on concentric rings to demonstrate generational depth. This methodology not only tackles the problems of standard graph techniques by reducing clutter and improving navigability, but it also expands the user's comprehension by incorporating biographical and geographical material. Furthermore, the tool is available via web and mobile applications. Our tool marks a big advancement in making the complex networks of Hadith narration more accessible and engaging for scholars and enthusiasts alike.

Acknowledgment. This work is part of a research project that is support by Qatar Research, Development and Innovation Council (QRDI) (formerly known as Qatar National Research Fund (QNRF)) under grand number: NPRP13S-0129-200194.

Disclosure of Interests. The authors have no competing interests.

References

1. Dimashqi, A.-H.I.K.: Al Bidaya Wa Al-Nihaya (The Beginning and the End). Dar Al-Kotob Al-Ilmiyah (2002)
2. Swarup, R.: Understanding the Hadith: The Sacred Traditions of Islam. Prometheus Books, Amherst, N.Y. (2002)
3. al Asqalani, I.H.: Nukhbat al-Fikar fi Mustalah Ahl al-Athar. Dar al-Kotob al-Ilmiyah (2002)
4. Khan, I.A.: Muslim scholars' contribution in hadith authentication. J. Islam Asia (E-ISSN 2289-8077) **8**, 345–363 (2011)
5. Saeed, S., Yousuf, S., Khan, F., Rajput, Q.: Social network analysis of hadith narrators. J. King Saud Univ.-Comput. Inf. Sci. **34**(6), 3766–3774 (2022)
6. Shukur, Z., Fabil, N., Salim, J., Noah, S.A.: Visualization of the hadith chain of narrators. In: Zaman, H.B., et al. (eds.) IVIC 2011. LNCS, vol. 7067, pp. 340–347. Springer, Heidelberg (2011). https://doi.org/10.1007/978-3-642-25200-6_32
7. Siddiqui, M.A., Saleh, M., Bagais, A.A.: Extraction and visualization of the chain of narrators from hadiths using named entity recognition and classification. Int. J. Comput. Linguist. Res **5**(1), 14–25 (2014)
8. Borg, I., Groenen, P.J.F.: Modern Multidimensional Scaling. SSS, Springer, New York (2005). https://doi.org/10.1007/0-387-28981-X
9. Van der Maaten, L., Hinton, G.: Visualizing data using t-SNE. J. Mach. Learn. Res. **9**(11) (2008)
10. McInnes, L., Healy, J., Melville, J.: UMAP: uniform manifold approximation and projection for dimension reduction. arXiv preprint arXiv:1802.03426 (2018)
11. Muslim scholar database. https://muslimscholars.info/. Accessed Jan 2023